

Stimulus Analysis: Choosing a Perception Radius and a Termination Threshold

How the dispersion stimuli are measured, how the perception radius was selected, and how clips are terminated so that conditions end at the same degree of dispersion.

1. The problem

Adding random movement makes a swarm disperse further as well as differently. If every clip runs for a fixed time, the randomised conditions end both more spread out *and* more erratic, and the two are confounded: a participant judging the stimuli cannot separate the manner of the motion from how far the swarm got.

The fix is to hold the endpoint constant. The run with no randomness disperses to a plateau under the separation rule alone. That plateau area becomes the stop condition for every other condition, so all clips end at the same spread and randomness changes only the character of the motion.

Two things must be true of the perception radius chosen for this. The swarm has to **settle**, rather than expand until the walls stop it, and it has to get there without **most of the swarm pressed against the walls**. A wall-limited plateau is not a property of the swarm, it is a property of the arena.

2. What is measured

Two series per frame, both computed in Unity at capture time and stored in the recording:

Measure	Where computed	Definition
Trimmed hull area	SwarmDensityMetrics.cs	Convex hull of the agents by Andrew's monotone chain, area by the shoelace formula, after discarding the 10% of agents furthest from the centroid. The trim stops a single straggler dominating the measure.
Wall contacts	SwarmTrajectoryRecorder.cs	Agents within radius + skin of a wall surface, tested only against the walls active for that run, so a motion type that disables walls is not credited with phantom contacts.

Both are recorded rather than recomputed downstream, so the analysis reads exactly the quantity the recorder was deciding on and there is a single implementation of each measure. The hull areas are independently checked against `scipy.spatial.ConvexHull` by `validate_hull.py`; agreement is at the $1e-5$ level, the residual being the three decimal places the recordings are written at.

3. Detecting a steady state

A plateau is not the last frame. A run still slowly expanding when the timeout arrives would look settled if the final value were taken at face value. Instead the hull slope is measured over a rolling window and the plateau is the first point from which the slope stays small for a sustained period:

```
slope(t) = [A(t) - A(t - w)] / w w = 1.0 s
settled = |slope| <= tol * max(A) tol = 0.02 per second
plateau = first index from which settled holds for 3.0 s
A* = median A over the plateau
```

Expressing the tolerance as a fraction of the run's own area scale means one setting works whether a swarm settles at $25 u^2$ or $900 u^2$. A run that never satisfies the condition is reported as `settled = False` and its final area treated as a lower bound rather than a plateau.

The wall budget

At the plateau index, the analysis reports how many distinct agents have touched a wall at any point, as a fraction of the swarm. The acceptable fraction is configurable and defaults to **20%**. Alongside it, the plateau area is expressed as a fraction of the usable arena (1009 u² for Scene6_Expand): a value near 1 means the arena set the limit.

4. Results: Dispersion

Reference runs are those with random movement 0. Measured over the 18-run Dispersion sweep, 40 agents, friction 0.90:

Perception	Max speed	Plateau	Settles at	Arena	Agents on walls	Within budget
0.15	1.5	24.9 u ²	1.0 s	2%	0 (0%)	yes
0.15	4.0	24.8 u ²	1.0 s	2%	0 (0%)	yes
3.15	1.5	390.1 u²	12.4 s	39%	4 (10%)	yes
3.15	4.0	406.1 u ²	7.0 s	40%	6 (15%)	yes
42.15	1.5	967.3 u ²	19.0 s	96%	39 (98%)	no
42.15	4.0	979.2 u ²	10.3 s	97%	35 (88%)	no

The three radii fall into three regimes, and only the middle one is usable:

0.15 does not disperse at all. The agents cannot perceive one another, so the separation rule has nothing to act on and the hull never leaves its spawn value.

42.15 saturates the arena. It reaches a flat line, but at 96% of the available area with 39 of 40 agents having touched a wall. The plateau is imposed by the geometry, not by the swarm reaching equilibrium, and it would move if the arena were resized.

3.15 settles genuinely. The hull stabilises at 390 u², 39% of the arena, with 4 of 40 agents having brushed a wall. This is a property of the separation rule at that perception radius.

Sensitivity

The choice is not an artefact of the settings. It holds for wall budgets of 10%, 20% and 30%, and for slope tolerances of 0.01, 0.02 and 0.05, the last shifting only the reported plateau time from 12.4 s to 9.0 s. At a 5% budget the recommendation collapses to 0.15, which is to say no radius disperses at all without some wall contact.

5. The threshold

```
end condition AbsoluteHullAreaReached
target 390 u2
dwell 0.5 s
max time 50 s (override, above the slowest firing time in section 6)
```

Expressed as an absolute area rather than a ratio to each run's own starting area. A ratio would also work while every Dispersion run spawns into the same footprint, but it would silently change meaning the moment a spawn area did. The direction of the test, grow-to versus shrink-to, is inferred from the area at recording start.

Why a dwell time

With randomness the hull oscillates; the random-80 traces swing by roughly $\pm 100 u^2$. A bare threshold crossing would end a clip on a transient spike, at a moment that is not representative of the state. Requiring the condition to hold continuously for 0.5 s removes that without materially delaying the stop.

6. Verification against recorded runs

Applying the rule offline to the existing recordings, with the same dwell logic Unity uses, shows when each clip would have ended. At perception 3.15 it fires in all six runs:

Random	Max speed	Fires	At	Fraction of the recorded clip
0	1.5	yes	14.6 s	0.24
0	4.0	yes	6.7 s	0.11
40	1.5	yes	14.7 s	0.24
40	4.0	yes	6.6 s	0.11
80	1.5	yes	37.3 s	0.60
80	4.0	yes	5.8 s	0.09

At perception 0.15 the target is unreachable in four of six runs, the swarm never exceeding $231 u^2$, which is a further reason that radius is unusable. At 42.15 every run fires within 3.7 to 16.9 s, but those clips are the wall-saturated ones.

The trade-off this design makes

Equalising the endpoint means the duration varies instead. At the chosen radius the firing times span **5.8 s to 37.3 s**, a spread of 31 s. Clip length is visible to participants and could carry information about the condition, so it should be reported alongside the results rather than left implicit. The alternative, fixing duration, would reintroduce the confound this design exists to remove.

7. Scope: where the method does and does not apply

The same analysis was run over the other motion types. It transfers cleanly only to Dispersion, and the reasons are worth stating rather than discovering later.

Motion type	Finding	Applies?
Dispersion	Three clean regimes, a genuine plateau at 3.15, target $390 u^2$.	yes
Densification	Contracts to $6-9 u^2$ at every radius tested, with no wall contact. Perception barely changes the endpoint, so there is nothing to select on; the largest radius wins trivially.	not useful
Flocking	Hull area is unstable as a measure here: the flock translates across the arena and leaves through the disabled walls, so the hull tracks a moving cluster rather than a spread. One run never settles at all.	no
Random	No zero-randomness run exists, because randomness is the behaviour. There is no reference condition to define a target from.	no

For Densification the equivalent question would be a contraction target rather than an expansion one, which the same `AbsoLuteHullAreaReached` rule supports by inferring direction. For Flocking a

translation-invariant measure such as radius of gyration or mean nearest-neighbour distance would be needed, since hull area conflates spread with position.

8. Reproducing this

```
cd Analysis
python3 steady_state.py ../Assets/SimulationRecordings/Combinations/Dispersion/
python3 steady_state.py ../Assets/SimulationRecordings/Combinations/Dispersion/ \
--target 390 --dwell 0.5
```

```
# or interactively, with the plots and a sensitivity pass
Notebooks/choose_perception.ipynb
```

--budget changes the acceptable wall-contact fraction, --slope and --hold the plateau criteria. The notebook additionally sweeps those settings to show whether the recommendation is stable.

9. Limitations

The sweep has three points. 390 u^2 is the best of 0.15, 3.15 and 42.15, which are orders of magnitude apart. A finer sweep at random movement 0 across roughly 1 to 12 would establish whether 3.15 is genuinely the best compromise or merely the best available. The analysis re-runs unchanged on that data.

The trimmed hull is discontinuous. When two agents sit at nearly the same distance from the centroid, which one the trim discards is decided by numerical detail, and if one is a hull vertex the area can jump by a few percent between frames. Observed once in 3703 frames. The dwell time makes a threshold robust to it, but it is a property of the measure worth knowing.

The plateau depends on the arena. A_c and the arena fraction are computed for the Scene6_Expand arena at 1009 u^2 . Resizing the arena changes which radii are usable.

Max speed interacts with friction. At friction 0.9 the sustainable speed is $\min(\text{maxSpeed}, F/f)$, so raising max speed beyond that ceiling does not make a swarm reach the target sooner. This is why the two max speed columns differ in timing more than in plateau.

10. References

Hénard, A., Peillard, É., Rivière, J., Kubicki, S., & Coppin, G. (2024). *Human perception of swarm fragmentation*. HRI '24, Boulder, CO.

See also Docs/SwarmDensityReference.pdf for the density measures, the connectivity threshold and where the constant 4.51 comes from.